exercise 1. Estimate the probability that 1000000 coin flips of fair coins will result in more than 501000 heads, using CLT.

Solution:

We consider X_1 that represent the flip of a single coin, X_1 is a Bernoulli variable defined by $X_1 \begin{cases} 1 & p=\frac{1}{2} \text{ if it is head} \\ 0 & p=\frac{1}{2} \to \text{tail.} \end{cases}$

Here we are repeating the flip 1000000 times.

So it is like we are considering the variable

$$X = \sum_{i=1}^{n} X_i \quad \text{with } X_i = Be(\tfrac{1}{2}) \quad X_i \text{ are all iid.}$$

$$n = 1000000.$$

We want to estimate $\mathbb{P}(X \geq 501000)$.

By the central limit theorem we have that given $X_1, \dots, X_n$ iid. and given $X = \sum_{i=1}^{n} X_i$ we have that

$$\frac{X - \mathbb{E}(X)}{Sd(X)} \xrightarrow[n \to \infty]{d} N(0,1).$$

where with $\xrightarrow{d}$ I meant convergence in distribution, i.e. convergence of the CdF.

In other words $\forall a, b \in \mathbb{R}$

$$\mathbb{P}\left[a \leq \frac{X - \mathbb{E}(X)}{Sd(X)} \leq b\right] \to \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{x^2}{2}} dx.$$

Consider $X_1 = Be(\frac{1}{2})$ $\quad \mathbb{E}(X_1) = \frac{1}{2}$ $\quad Var(X_1) = \frac{1}{4}$

$$\to \mathbb{E}(X) = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}(X_i) = n \cdot \mathbb{E}(X_1) = \frac{n}{2}$$

($\sum$: $\downarrow$ independent; $\mathbb{E}$: $\downarrow$ iid.)

$$n = 1000000 \Rightarrow \mathbb{E}(X) = 500000$$

$$Var(X) = Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) = n\, Var(X_1) = \frac{n}{4} \to Var(X) = 250000$$

($\sum$: $\downarrow$ independent; Var: $\downarrow$ iid)

$$\Rightarrow Sd(X) = \sqrt{250000} = 500$$

By CLT

$$\mathbb{P}(X > 501000) = \mathbb{P}\left(\frac{X - 500000}{500} > \frac{501000 - 500000}{500}\right)$$

$$= \mathbb{P}\left(\frac{X - 500000}{500} > 2\right) \sim \mathbb{P}(N(0,1) > 2).$$

We have $\mathbb{P}(N(0,1) \leq 2) = 0.9772 \Rightarrow \mathbb{P}(N(0,1) > 2) = 1 - 0.9772 = \boxed{0.228}$

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exercise 2. $\Omega=[0,1]$. P uniform probability in Ω

$\Rightarrow$ the density function $p(x)=\begin{cases}1 & \forall x\in[0,1]\\ 0 & \text{otherwise.}\end{cases}$

$X_1,\dots,X_n$ independent uniform ~~r.v.~~ r.v., $f:[0,1]\to\mathbb{R}$.

Show that $$\lim_{n\to+\infty}\mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n}f(X_i)\right)=\int_0^1 f(t)\,dt$$

$$\lim_{n\to\infty}\mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n}f(X_i)\right)=0$$

↳ Solution:

Consider $S_n=\frac{1}{n}\sum_{i=1}^{n}f(X_i)$, since the expectation is linear.

$$\mathbb{E}[S_n]=\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[f(X_i)]$$

$X_i\sim U([0,1])$, by definition $\mathbb{E}[X_i]=\int_{\mathbb{R}} t\,f_{X_i}(t)=\int_0^1 t\,dt$

(f_{X_i}: pdf of X_i)

and $\mathbb{E}(f(X_i))=\int_0^1 f(t)\,dt$ — If is the same for every i

Then $\mathbb{E}[S_n]=\frac{1}{n}\sum_{i=1}^{n}\int_0^1 f(t)\,dt=\int_0^1 f(t)\,dt \quad \forall n.$

So if we take the limit as $n\to+\infty$ $\lim_{n\to+\infty}\mathbb{E}[S_n]=\int_0^1 f(t)\,dt.$

Now we consider

$$\mathrm{Var}(S_n)=\frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n}f(X_i)\right)=\frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(f(X_i))=\frac{1}{n^2}\cdot n\cdot\mathrm{Var}(f(X_1))$$

(same distribution)

$$=\frac{\mathrm{Var}f(X_1)}{n}$$

↳ [They are independent.
Theo: If X,Y are indep. r.v. and f,g are continuous functions $\Rightarrow f(X)\perp\!\!\!\perp g(Y)$ are independent].

$\mathrm{Var}f(X_1)$ is a finite value

$\Rightarrow$ taking the limit $\lim_{n\to+\infty}\frac{\mathrm{Var}f(X_1)}{n}=0.$

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exercise 3 $X=(X_1,\dots,X_n)$ random sample size n from a Binomial distribution with parameters n and p. $X_1 \sim Bin(n,p)$

$\mathbb{E}[X_1] = np$

$Var(X_1) = np(1-p)$.

a) Compute Fisher information $I_X(p)$.

Given a family $\{f_\theta \mid \theta \in \Theta\}$, X random vector with distribution f_θ

the Fisher information is defined as: $I_X(\theta) = \mathbb{E}_\theta\left[\left(\frac{d}{d\theta}\log f_\theta(X)\right)^2\right]$, $\forall \theta \in \Theta$

First consider the Joint density function of $X=(X_1,\dots,X_n)$.

$$f_X(K) = \prod_{i=1}^n f_{X_i}(k_i) = \prod_{i=1}^n \binom{n}{k_i} p^{k_i}(1-p)^{n-k_i} \qquad K=(k_1,\dots,k_n).$$

$$\ln f_X(K) = \log\left(\prod_{i=1}^n \binom{n}{k_i} p^{k_i}(1-p)^{n-k_i}\right)$$

$$= \sum_{i=1}^n \log\binom{n}{k_i} + \sum_{i=1}^n \log\left(p^{k_i}\right) + \sum_{i=1}^n \log(1-p)^{n-k_i}$$

$$= \sum_{i=1}^n \log\binom{n}{k_i} + \sum_{i=1}^n k_i \log(p) + \sum_{i=1}^n (n-k_i)\log(1-p)$$

$$= \sum_{i=1}^n \log\binom{n}{k_i} + \left(\sum_{i=1}^n k_i\right)\log(p) + n^2\log(1-p) - \left(\sum_{i=1}^n k_i\right)\log(1-p)$$

Define $S = \sum_{i=1}^n X_i \Rightarrow \ln f_{p(x)} = \sum_{i=1}^n \log\binom{n}{k_i} + S\cdot\log(p) + n^2\log(1-p) - S\log(1-p)$.

we have $\frac{d}{dp}\ln f_p(X) = \frac{S}{p} - \frac{n^2}{1-p} + \frac{S}{1-p} = \frac{S(1-p) - n^2p + Sp}{p(1-p)}$

$$= \frac{S-n^2p}{p(1-p)}$$

$$\Rightarrow I_X(p) = \mathbb{E}_p\left[\left(\frac{S-n^2p}{p(1-p)}\right)^2\right] = \frac{1}{p^2(1-p)^2}\mathbb{E}_p[(S-n^2p)] = \frac{n^2p(1-p)}{p^2(1-p)^2} = \boxed{\frac{n^2}{p(1-p)}}$$

$$\mathbb{E}_p[S] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = n\cdot\mathbb{E}[X_1] = n(np) = n^2p$$

$$\Rightarrow \mathbb{E}[S-n^2p] = Var(S) = nVar(X_1) = n^2p(1-p)$$

Alternatively since X_i are iid we know that $I_p(X) = nI_p(X_1)$ (so we could have use the binomial density function)

b) Z unbiased estimator of p^2. Cramér-Rao inequality for Z.

$\mathbb{E}(Z) = p^2$

The Cramér-Rao inequality states that given $Y = t(X)$, $g_\theta = \mathbb{E}_\theta(Y)$

$$\mathrm{Var}_\theta(Y) \geq \frac{|g'(\theta)|^2}{I_X(\theta)} \quad \forall \theta \in \Theta$$

Here. $Z = t(X)$

$\mathbb{E}_p[Z] = p^2 = g(p) \rightarrow g'(p) = 2p \rightarrow |g'(p)|^2 = 4p^2$

$I_X(p) = \frac{n^2}{p(1-p)}$

C-R inequality is $\mathrm{Var}(Z) \geq \frac{4p^2}{\frac{n^2}{p(1-p)}} = \boxed{\frac{4p^3(1-p)}{n^2}}$

c) W unbiased estimator of $\frac{1}{p}$. Cramér-Rao inequality for W.

$g(p) = \frac{1}{p} \rightarrow g'(p) = -\frac{1}{p^2} \rightarrow |g'(p)|^2 = \frac{1}{p^4}$

C-R inequality is $\mathrm{Var}(W) = \frac{\frac{1}{p^4}}{\frac{n^2}{p(1-p)}} = \frac{1}{p^4} \cdot \frac{p(1-p)}{n^2} = \frac{1-p}{p^3 \cdot n^2}$

Exercise 4 $X_1, \dots, X_n$ sample of size n, $X_i \sim Po(\lambda)$ with $\lambda > 0$.

$\mathbb{P}(X_i = k) = e^{-\lambda} \frac{\lambda^k}{k!} \quad \forall k \geq 0.$

a) Find MLE for λ.

b) Is the MLE* unique?

Solution: To find the MLE we need the likelihood function $L(\lambda)$ which is the joint probability function.

$$L(\lambda) = \prod_{i=1}^{n} f_{X_i}(k_i) = \prod_{i=1}^{n} \frac{\lambda^{k_i}}{k_i!} e^{-\lambda} = \frac{e^{-n\lambda} \prod_{i=1}^{n} \lambda^{k_i}}{\prod_{i=1}^{n} k_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^{n} k_i}}{\prod_{i=1}^{n} k_i!}$$

↓ product cause X_i are iid.

We want to maximize $L(\lambda)$ wrt λ, this is the same as maximize the $\log L(\lambda)$

$$\log L(\lambda) = \ln L(\lambda) = \ln\left(e^{-n\lambda} \cdot \lambda^{\sum_{i=1}^{n} k_i}\right) - \ln\left(\prod_{i=1}^{n} k_i!\right)$$

$$= -n\lambda + \left(\sum_{i=1}^{n} k_i\right) \ln(\lambda) - \sum_{i=1}^{n} \ln(k_i!)$$

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To find the max we consider the derivative and set it to zero.

$$\frac{d}{d\lambda} \log \mathcal{L}(\lambda) = -n + \frac{\left(\sum_{i=1}^{n} x_i\right)}{\lambda} = 0$$

$$\left(\sum_{i=1}^{n} x_i\right) = n\lambda \rightarrow \lambda = \frac{\sum_{i=1}^{n} x_i}{n}$$

Then the estimator is $\hat{\lambda} = \frac{\sum_{i=1}^{n} X_i}{n} = \overline{X}$

Thus $\hat{\lambda}$ is the minimum, consider the second derivative.

$$\frac{d^2}{d\lambda^2} \log(\mathcal{L}(\lambda)) = -\frac{\left(\sum_{i=1}^{n} x_i\right)}{\lambda^2}$$

$x_i \geq 0 \ \forall i$, $\lambda > 0$ $\rightarrow$ $\frac{d^2}{d\lambda^2} < 0 \quad \forall \lambda > 0$ so the function is strictly concave $\Rightarrow$ the max is unique.

Even if x_i are all zero $\mathcal{L}(\lambda) = -n\lambda < 0$ strictly decreasing.
$\Rightarrow$ unique sup at zero
so the estimator in this case is $\hat{\lambda} = 0$ that is the sup.